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Are Emojis the New Words? A Sentiment Analysis of Social Media Brand Conversations

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Are Emojis the New Words? A Sentiment Analysis of Social Media Brand Conversations

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ABSTRACT

Emojis have become an increasingly important aspect of consumer-brand interactions in the Indian subcontinent. However, the impact of emoji use on brand image and mental health remains underexplored, particularly in emerging economies like India, where structured research on this topic is limited. To address this gap, the present study analyzes over 4,600 consumer tweets related to 19 prominent brands across eleven industries. Using VADER sentiment analysis, the research develops a metric to assess consumer sentiment and brand engagement in relation to emoji usage. The findings indicate that effective integration of emojis contributes to positive consumer sentiment and enhanced brand engagement. Given that emojis have become a fundamental part of everyday communication, this study recommends that brand and marketing managers closely monitor consumer interactions with emojis, utilizing rich data to refine communication strategies and maximize emoji-driven engagement.

Keywords: Emoji, Consumer Behavior, Sentiment Analysis, Social Media, Marketing, VADER

INTRODUCTION

Over time, emojis have evolved from simple pictographs into a crucial component of modern communication, marking a significant shift in the way individuals and brands interact within the digital and social media landscape. Their use is particularly prevalent in India's intricate and diverse market, where emojis have become instrumental in navigating the country's rich linguistic and cultural diversity. In India's diverse and multilingual market, emojis serve as a universal communication tool that helps brands engage effectively with consumers across different linguistic and cultural backgrounds.

This study examines how emojis influence brand equity and consumer engagement in India by analyzing sentiment in social media interactions. Using VADER sentiment analysis on tweets related to prominent Indian brands, we explore whether emojis function as mere decorative elements or strategic marketing tools.

In the dynamic and competitive Indian market, where brands vie for attention across diverse cultural and linguistic communities, the use of emojis offers a way to transcend the limitations of language. India's linguistic diversity, with over 22 official languages and numerous dialects, often makes traditional text-based communication a challenging endeavor for marketers. Emojis, however, function as a common visual language that can be easily understood by a wide audience, regardless of their linguistic background. This makes them particularly valuable in a country like India, where a brand's ability to connect with consumers across regional and linguistic divides is critical to its success. For instance, emojis can convey emotions, moods, and ideas in a way that text alone often cannot, enabling brands to communicate more effectively with their target audiences.

Gen Z, the first digitally native generation, prefers fast and visually expressive communication. Emojis align with their engagement style, making them a crucial tool for brands aiming to connect with younger audiences on social media.

In India, where digital engagement is rapidly increasing among younger consumers, the strategic use of emojis has become a powerful marketing tool. Brands such as Swiggy, Zomato, and Flipkart have already recognized the potential of emojis to create more engaging and relatable content, often using food and shopping-related emojis to evoke emotions such as hunger, excitement, or satisfaction. These brands have effectively utilized emojis to create personalized and fun interactions with consumers, which not only drives engagement but also

reinforces positive brand associations. The emotional appeal of emojis, combined with their ability to cut across linguistic barriers, allows brands to communicate more effectively in a market as diverse as India.

This study, therefore, examines how the integration of emojis within brand messaginghas evolved into a thoughtful and deliberate practice in Indian marketing. As the digital landscape continues to grow in complexity, and as consumer expectations evolve, the use of emojis has become a strategic tool for brands to enhance their emotional connection with consumers. By analyzing the sentiment in tweets related to major Indian brands, this research aims to demonstrate how emojis contribute to the expression of consumer sentiments, shape brand perceptions, and ultimately foster brand loyalty. The study also highlights how the growing ubiquity of emojis among younger consumers, particularly Gen Z, is influencing the way brands communicate, with emojis becoming a central element in the development of brand equity in the competitive Indian market.

The central question this study seeks to address is:

"How does the use of emojis in social media conversations impact consumer sentiment and brand equity in the Indian market?"

In addition to our primary question—'How does the use of emojis in social media conversations impact consumer sentiment and brand equity in the Indian market?' —this study now also examines three specific dimensions: emoji type (positive, negative, neutral), usage context (e.g., promotional, customer service, organic conversation), and frequency. We hypothesize that tweets with a higher proportion of positive emojis yield higher sentiment scores, that the impact of emojis varies by context, and that increased emoji usage correlates with greater consumer engagement and enhanced brand equity.

In conclusion, the increasing use of emojis in digital marketing reflects a broader shift toward more visual, emotion-driven communication. As brands in India seek to differentiate themselves in a crowded marketplace, the ability to create meaningfuland memorable interactions with consumers has become critical. Emojis offer a unique opportunity for brands to stand out, capture attention, and create deeper emotional connections with their audiences. This research, by focusing on sentiment analysis of emoji usage in tweets related to Indian brands, aims to contribute to the growing body of knowledge on digital marketing and

consumer behavior, particularly in the context of India's diverse and rapidly evolving market.

LITERATURE REVIEW

Emojis have become an essential part of digital communication, enriching emotional expression and social interaction in text-based platforms where traditional non-verbal cues are absent. Early research by Derks et al. (2007) established that emoticons, the predecessors of modern emojis, clarify emotional intent and strengthen interpersonal connections in socially rich online environments. Expanding on this foundation, Kaye and Schweiger (2023) investigated whether emojis serve as reliable indicators of users' immediate emotional states, finding that on platforms like Twitter, they effectively mirror in-the-moment moods. While Derks et al. (2007) highlight thebroader social dynamics of emoticon use, Kaye and Schweiger (2023) focus on real-time emotional reflection, though both underscore emojis' role in compensating for the emotional limitations of text.

However, these studies predominantly examine Western contexts, leaving a significant gap in understanding how emojis facilitate emotional expression in culturally diverse settings like India. Cultural factors play a critical role in shaping emoji usage, as evidenced by Alzara and Mellor (2024), who identified distinct emoji preferences in the UAE that reflect local communication norms, and Daud and McLellan (2016), who analyzed Bruneian Facebook posts to reveal how emojis intertwine with gender and language to express identity. Although both studies emphasize cultural specificity, Alzara and Mellor (2024) explore general usage trends, while Daud and McLellan (2016) delve into demographic details, yet neither provides a cross-cultural comparison, particularly for multilingual nations like India where emojis might serve as a unifying tool across linguistic divides.

In the marketing domain, emojis have emerged as powerful tools for enhancing brand engagement on social media. Ballester et al. (2023) demonstrated that emojis increase user interaction with eco-friendly restaurants on Instagram, reinforcing brand identity and trust, while Trivedi and Singh (2021) found that positive emojis in consumer reviews of Indian food delivery apps correlate with heightened satisfaction and loyalty.

Although both studies confirm emojis' positive influence, Ballester et al. (2023) emphasize visual branding, whereas Trivedi and Singh (2021) focus on consumer sentiment, yet neither isolates emojis' specific impact on brand equity, especially in diverse markets like India where cultural variety could magnify their strategic importance. Sentiment analysis provides a structured approach to decoding these consumer emotions in digital spaces, with Trivedi and Singh (2021) linking emoji-inclusive reviews to brand sentiment in food delivery apps, and Kaye and Schweiger (2023) validating emojis as mood markers. These findings converge on emojis' ability to signal emotions—positive symbols like hearts indicate satisfaction, while negative ones like frowns suggest dissatisfaction. However, much of this research integrates emojis into broader text analysis, overlooking their standalone emotional significance, particularly in multilingual contexts like India where their meaning might differ across language groups.

Theoretically, emojis' influence on consumer behavior can be understood through prospect theory (Kahneman & Tversky, 1979), which posits that individuals assess outcomes as gains or losses relative to a reference point. Positive emojis may frame brand experiences as gains, encouraging loyalty, while negative emojis signal losses, potentially reducing engagement. Though not originally applied to emojis, this framework offers a fresh perspective for marketing applications, yet its use in emoji research remains largely unexplored, marking a gap in connecting behavioral economics to digital communication strategies. Additionally, the emoji lexiconevolves alongside user needs, as Feng et al. (2020) showed by examining Twitter requests for new emojis, revealing how cultural shifts drive demand for new symbols that offer brands novel engagement opportunities. However, there is little research on how these new emojis are adopted in diverse cultural settings like India, where rapid digital growth might foster distinctive usage patterns.

Synthesizing these insights reveals several research gaps: studies like Alzara and Mellor (2024) and Daud and McLellan (2016) focus on specific regions but lack comparative analyses across diverse cultures, particularly India's multilingual landscape; research often prioritizes Western or select non-Western contexts, neglecting India's unique socio-cultural dynamics; while Ballester et al. (2023) and Trivedi and Singh (2021) explore engagement, they do not examine how emojis influence brand equity in varied markets; sentiment analyses frequently combine emojis with text, missing their independent emotional impact, especially in multilingual settings; the application of frameworks like prospect theory to emoji use in marketing is underdeveloped; and Feng et al. (2020) address emoji evolution, but adoption patterns in diverse cultural contexts remain unstudied. This research responds to the call for culturally diverse research noted by Bai et al.

(2019) and provides valuable insights for both academic and practical approaches to digital branding.

METRIC USED FOR THE SENTIMENT ANALYSIS

Here, we employ the VADER (Valence Aware Dictionary and sEntiment Reasoner), a tool enabling us to analyze emotional aspects of textual communication, especially in terms of emotion-bearing pictures, called emojis. They constitute a quicker way of searching the sentiment of micro blog and are not limited to words but also make use of emojis and other expressions. Consequently, it is particularly useful in analyzing texts of modern ways of interaction. In the subsequent sections, we shall explain how the VADER works including its calculations, and whether it is appropriate in the analysis of such types of texts full of emojis.

Overview of the VADER Model

It is the combination of sentiment lexicon and emotion-sensing heuristics that distinguishes VADER's sentiment analysis. It features a precompiled list of words annotated for sentiment and each word’s value ranges from -4 to 4. One of the main differences between VADER and standard approaches incorporating various dictionaries is that VADER uses such features of social networks as emoticons and emojis, punctuation symbols and capital letters. In this case, VADER is best suited for the examination of non-formal and condensed chunks of texts where such non-verbal signs are prevalent.

Key Components of Sentiment Analysis

VADER calculates sentiment scores based on several key components that capture the different shades of sentiment expressed in a given text. The positive sentiment score reflects the proportion of the text that conveys a positive emotional tone, calculated by summing the positive values assigned to individual words or emojis from the lexicon and normalizing this sum by the overall length of the text. Words such as "amazing," "great," and positive emojis like "😊" contribute positively to this score.

Conversely, the negative sentiment score indicates the degree of negativity present in the text. This score is calculated similarly to the positive score, summing words or emojis with negative values—words like "terrible," "bad," and negative emojis like "😢." Additionally, the neutral sentiment score captures the portion of the text

that lacks strong emotional charge, such as function words ("the," "is") or neutral statements ("It is raining"). VADER determines this score by measuring the proportion of the text that does not significantly influence the overall sentiment.

The compound score, a key metric used to assess the overall sentiment of the text, integrates all the aforementioned components into a normalized value between -1 (extremely negative) and +1 (extremely positive). VADER computes the compound score using a weighted average of the positive, negative, and neutral components, normalized to fit within this range. Mathematically, the compound score can be represented as:

$$Compound = \frac{\sum(Polarity\ Scores)}{\sqrt{\sum(Polarity\ Scores^2) + \alpha}}$$

Here, α is a small smoothing constant that prevents division by zero errors and ensures that very small compound values are handled appropriately.

Heuristics and Rule-Based Enhancements

One of the most interesting things about VADER is how it employs heuristics in order to modify sentiment scores across texts according to language structure and usage. Rules like those mentioned above take care of tonal differences that may be lost by unnatural usually lexicon dependent abilities. Thus, VADER based on the punctuation further categorizes and scales sentiment – the more exclamations in the text the more passionate emotion involved. For instance, “great!!!” would produce a much greater positive sentiment than just “great,” whereas “terrible!!!" would be a much stronger negative sentiment than, “terrible.”

Capitalization also plays a role; capitalized words are assigned greater weight in the sentiment score as they often indicate emphasis. Thus, "AMAZING" would score higher positively than "amazing." Additionally, degree modifiers (or intensifiers) suchas "very," "extremely," or "kind of" adjust the polarity of words. For instance, "very happy" scores higher on the positive scale than just "happy," while "kind of sad" lowers the negative score compared to "sad." VADER effectively handles negation by reversing the polarity of sentiment; for instance, "not happy" would negate the positive score associated with "happy," ultimately decreasing the overall sentiment.

VADER also interprets emojis and emoticons similarly to words. Positive emojis, like "😊," contribute to the positive sentiment, while negative emojis, such as "😢," contribute to negative sentiment. This feature is crucial for our project, given

that emojis represent a significant form of non-verbal emotional expression in modern digital communication.

Handling Contextual Sentiment

One of the challenges in sentiment analysis, particularly with emojis, is capturing sentiment that extends beyond the literal meanings of words. VADER is adept at interpreting the sentiment of short, context-specific texts typical of social media platforms, making it especially suitable for analyzing emoji usage. For instance, in the sentence "I'm not sure 😐," the neutral emoji "😐" complements the uncertainty expressed, resulting in an overall neutral sentiment. Whereas, "I love it! 😊" would yield a strong positive score due to the combined effects of both the positive words and the positive emoji.

Advantages of VADER for Emoji Sentiment Analysis

Specifically within this paper's scope, VADER offers various important advantages over other approaches when performing sentiment analysis with symbols such as emojis. Its lexicon is designed for the spoken rather than written language because of the peculiarities of informal text communication. This tailoring enhances the effectiveness of VADER in social media outlets which are more embedded with emojis. More so, English has other gibberish languages that VADER understands and this provides maximum sentiment information from the text, which is very useful in trying to understand the usage of emojis specifically in India.

VADER is also designed to perform well in short text parsing, which is the case for social media such as X, WhatsApp and Instagram which incorporates frequent use of emojis. Accuracy thus enables classification of the sentiments through the analysis of forms of language usage that is informal and often, incomplete. In addition, VADER also allows simple interpretation of the output that is positive, negative, neutral and compound for all information provided in text form hence perception on sentiment from different texts becomes easier.

Limitations and Considerations

Strengths notwithstanding, VADER has its own drawbacks as well. Although effective in informal communication, it may face challenges in informal contexts aswell. More vague communicative forms, including sarcasm and irony, and literature, for example, may also prove to be difficult as they employ culture-based concepts, for example, emoticons. Whereas it can process negations and degree modifiers appropriately, it seems that VADER may not appreciate the sentiment in too stylized or sophisticated language.

Further, although VADER is able to operate on a total of 3,570 emojis, only 625 emojis are made available on the Dabbas (2023) emoticon's database that VADER is able to recognize and analyze sentiments. This limitation implies that quite a number of popular and frequently used emojis in the present word VADER might fail to capture effectively which might reduce the effectiveness of sentiment analysis as regards more uncommon or newish emoji. Hence, this limitation indeed is likely to affect the degree of sentiment analysis achieved in content that is characterized by the frequent use of emojis. In conclusion, VADER represents a robust way of analyzing sentiment of texts in a digital format with an emphasis on emoji usage. In this case, the bridled multilingual and emotive comprehension oils of the vocabulary allows quick and precise grasp of the ideas expressed by contemporary communication. However, certain aspects of VADER may limit the research and practice of digital sentiment analysis.

METHODOLOGY

X (formerly known as Twitter) as a data source:

As X (formerly Twitter), the platform offers considerable value to this research due to its historical use of emojis. An estimated 26.7% of the tweets analyzed between 2013 and 2023 contained emojis, demonstrating its utility for sentiment analysis (Dixon, 2023). With a long-standing role in conveying digital sentiments and emotional expressions, X presents a rich source for studying the emotional dynamics of online communication, making it a phenomenon worthy of deeper exploration.

Away from emoji usage, X is also used as a channel of advertising, today where brands participate in an active dialogue to create their brand image and to react to brand's audience (Culotta & Cutler, 2016; Leung & Chan, 2017). It facilitates understanding the brand from the customer's perspective and the customers' emotional response toward the brand.

As of May 2024, X is one of the most popular microblogging sites in the world, with more than 600 million active users monthly and is characterized by various user-content creation. Many of the accounts are public, and therefore many researchers do not need to solicit participation as there are few ethical concerns about privacy, which holds true for marketing and social media situation research (Culotta et al., 2016; Lambrecht et al., 2018).

Data Collection

The data for this research study was very carefully extracted in the month of September 2024 with the aid of a web scraping application that was developed using Python's BeautifulSoup package. It was considered so as to avoid any ambiguity, while extracting data regarding emoji usage in this case among tweets. Even though tools such as the twitteR package (Gentry, 2015) have been developed for the purpose of collecting tweets, the flexibility from using a custom scraper ensured that the data collected was of optimal quality specific for the objectives of this research.

In addition to the technical aspects of our data collection, the research was conducted with consideration of Twitter's (now X's) policies and ethical guidelines. Our data scraping process aimed to utilize publicly available information while being mindful of Twitter's Developer Agreement, rate limits, and user privacy standards. Efforts were made to limit the collection of personally identifiable information to what is openly accessible, and procedures were conducted with institutional ethical standards in mind. Our research process took into account various ethical and policy considerations, contributing to the overall integrity and reproducibility of our findings.

The research was done on 19 major B2C companies which operate within nine industries in India that are online and lamanating with campaigns especially on X (twitterT). The selected brands range from telecommunications and e-commerce to food delivery and lifestyle and explains how different types of emojis are used by the consumers while interacting with brands from diverse sectors in India.

Every brand selected was chosen for its robust digital presence, strong consumer connection, and strategic representation within its industry. By including brands from diverse sectors—such as telecommunications, dairy, e-commerce, and food delivery—we capture a broad spectrum of consumer interactions reflective of India's cultural, economic, and linguistic diversity. The selection was based on three key criteria: the brand's overall popularity, its prominent presence on Twitter, and its capacity to effectively represent the industry it belongs to. This multi-criteria approach ensures that our analysis is not confined to a single niche; rather, it provides a comprehensive view of how emojis function across varied contexts in India. As these brands actively target and engage with a young, digitally savvy demographic, including Gen Z, our study's findings can be confidently generalized to represent the dynamic landscape of Indian culture and consumer behavior.

To achieve the aims of the study, the researchers searched for X accounts that featured twice within the brand mentions, it was established that the database consisted of real consumers. For supplementary statistics, approximately 250 tweets (Refer Appendix A for detailed distribution of tweets across Brands) for each brand were retrieved focusing on tweets with emojis, especially because it was essential to analyze under the influence of how emojis affect the communication between the brands and theconsumers. To avoid incongruities in the data, tweets that included non-conventional symbols apart from the fixed emojis were excluded for example, emoticons or Japanese symbols. These procedures made it possible for emblems and the like to be used in such that there are no extraneous or irrelevant images in the comments (Peterson, 2012). It was aimed at investigating the use of emojis in digital communication with regard to emotions and images in interpretation, particularly how such symbols help for potential positional change of brands in close proximity to Gen Z consumers who are predominantly visual and digital.

The 19 brands included in the study cover eleven important sectors and represent a fair portion of the Indian consumer market. These include communication, online shopping, food ordering, edtech, fashion and lifestyle, transport, e-commerce and grocery, banking, entertainment, Dairy, Aviation, and Fintech. Each of these sectors is significant in the lives of Indian consumers with special emphasis on the Gen Z who demands interaction with brands that are quick, tailored, and emotionally relevant. The authors have built in the brands, which largely represent the deprived of aspirational consumption beginning from the basic requirements of telecom services to aspirational fashion and lifestyle segments. There are no such studies, therefore, which provide insights on how emojis can enrich communications for different customer segments with special emphasis on gen Z communication preferences.

Aviation: Air Vistara

Within the scope of its business, Air Vistara is the most customer centric and service oriented airline. As an apparently deficient premium carrier, the company primarily targets that does not tolerate either being uncomfortable for hours or being without services – the young professional travelers, and Gen Z. The application of emoji in the modes of communication used by Air Vistara allows conciseness and emotions related to travel in a more articulative manner. For example, the use of airplane emojis can be an excitement about travel, while that of a smiling face is about customer satisfaction. The way Air Vistara uses social media networks is most appealing to young people, the target audience the carrier

is interested in as well as the carrier's adaptation to the cultural context of Gen Z. The effect of this strategy is not only to win young travelers but also to secure the position of Air Vistara as a customer oriented airline carrier.

Dairy: Amul

Amul has been successful in carving a niche in the dairy industry and is a brand that is well known to the masses for its dairy offerings that form a part of millions in the daily diets of Indians. The emblem of the brand in its marketing element contains the emoji, which is of help to the Gen Z audience as it carries great significance. Emojis, are quite useful to this generation in that they offer nostalgic Fun images of the brand's past history and its associated values. With regard to imprinting product ideas in the process, the two combinations relate quite well whereby a milk emoji can represent the freshness and the desirability of the product while a heart one can indicate the love for the brand's products. Effects in the use of emojis include building up the brand image and bringing in loyalty amongst the Amul consumers. Inserting an emoji in a sensible strategy also shows that Amul is focused on maintaining high quality, yet knows how to approach an audience of children and teenagers.

E-Commerce: Flipkart, Myntra, TataCLiQ

The E-commerce App Flipkart, Myntra, TataCiq, etc. has not only revolutionized the shopping experience but has also increased the addiction level of shopping in the Indian population and in a much quicker span of time than expected. These brands target wide array of customers and provide them with the products such as electronics, fashion, groceries, household items, and so on. Though, for the gen Z, E-commerce is not only a medium to shop; it is an air around which certain experiences evolve, something like emersive participation. Tweets regarding these brands usually have emojis portraying excitement, a sense of an urgency to buy, satisfaction and so on especially during sales or product launches. For example a shopping bag emoji or fire emoji tries to capture the attention of the audience to grab limited-time deals, which plays well into Gen Z's need for quick and special offers. These consumers are used to being bombarded with visual and active content. Because of this, emojis enhance the promotional activities of retailers like Flipkart, Myntra and Tata CLiQ making them more memorable. By use of emojis in the communication of proper brands, e-commerce ensures that emotions are expressed more in the communication. This helps brands maintain an active and youthful image that belies the Gen Z's lifestyle.

Education & Technology: BYJUS, CRED

Both education and financial technology represented by brands like BYJUS and CRED optimize on unique customer groups like students and credit card holders respectively. Noting that these brands target younger consumers, Gen Z who is quite rich in technology and appreciates both education and financial knowledge. For younger learners, BYJUS used among other strategies emojis that brought fun into learning by making children influenced by the content more active. A CRED digital platform in a much similar manner incorporates the use of 'WOW' emoticons for users who are often presented with financial education. Emotions are critical to Gen Z as they do not feel that the traditional norm in economics and education in educational and financial centers is very easy with structures. Such emojis are explained to be used to capture the attention of the consumers at the same time making them understand the complicated matters such as that of the education system and the finance system targeted to millennials.

Entertainment: Dream11

Dream11 is one of the top fantasy sports sites that realize the love of sports, especially among Gen Z which includes the youngest enthusiastic cricket and other sportsfollowers. Emojis in Dream11 tweets are about thrill, rivalry, and fun, the core concepts for young sports lovers, the fans making crickernets crazy. It will be less difficult for Gen Z who love digitally interactive and competitive experiences to appreciate ie emojiies such as a trophy, fire or sports equipment to connect with the brands communication. These help to bulk up the aggression that accompanies play when involved in fantasy sports hence making the brand communication in Dream11 much fun. It is suggested that Dream11 branding in such cases should be moderated using these visual icons in order for the Gen Z sports audience to fully embrace their brand and passion for sports.

Fashion & Lifestyle: Lenskart, Tanishq, Nykaa

Consumers interested in fashion and lifestyle brands such as Lenskart, Tanishq and Nykaa tend to be more self expressive and focused on the aesthetics which matters particularly to the Gen Z. These brands deploy emojis to augment their products not only through pictorial representations of their goods such as sunglasses or diamonds to symbolize new eye wear and luxury exclusivities respectively. Such psychological advantages are especially useful for Gen Z whose brand ideology revolves around identity and self expression for planologists these emojis have helped in promoting direct brand interaction making it more personalized and

within the audience's ethos. Especially Nykaa, as a company engaged in the sale of beauty products, actively uses beauty and glamour emojis that serve to strengthen the brand as a trendsetter in the eyes of young fashionable consumers. All of these brands have succeeded in enhancing the brand experience comprising mainly the use of emoticons which is usually up to the expectations of the Gen Z who are modern consumers.

Food Delivery: Swiggy, Zomato

As either of the VCs or in-house specialists, visiting the operational locations of Swiggy and Zomato, two of the greatest food delivery companies in India, would envision a brand new service in time for a generation which cannot be waited for food reaching them. For this generation, which is instant and on-the-go, food delivery applications are lifestyle apps. Such emotions are embedded in the brand's communication and especially in the use of emojis. For instance, a pizza image or a heart that has eyes can create anticipation and craving very quickly connecting users to them. Since Gen Z is characterized by the use of images rather than texts, they will appreciate these emoticons; hence, the brand will appeal to them. Emojis help enhance customer support by making the interactions with representatives seem less practical, and more personable, which is very appealing to the younger generation. When communicating with their clients for example Gen Z, who regards service delivery speed and emotions highly, it is critical for Swiggy and Zomato to communicate through strategies they are comfortable with like emojis.

Fintech: Paytm

One company in the fin tech space, particularly in the sphere of electronic transactions and financial services and India, Paytm has not been a small player. The brand retails its products and services mainly to younger generations like Gen Z, who are always on the clock and need quick and hassle-free transactions. Young audience of the platform interestingly find the application of bitmojis by Paytm to market their services. As for example, some emoticons representing a wallet are used when services or products are paid for and others represent thumbs up when users want to show they trust the provider. Readers might observe that some marketing efforts and CRM in Paytm are targeted towards women and young people. This approach not only removes obstacles posed by complicated definitions of various financial terminologies but also creates a sense of trust and belonging, which makes it easy for the young generation portraying GenZ to adopt the use of electronic transactions in their active life.

Retail & Grocery: BigBasket, UrbanCompany

Brands such as BigBasket and UrbanCompany, target busy urban working professionals in the retail and grocery sector. These brands often deploy such emojis in advertisements to evoke fresh and instant aspects of the products, which is ideal for packs of Gen Z. For instance, there's some healthy and fresh produce indicated by a basket or leaves & veggies symbol in a BigBasket tweet whereas UrbanCompany may promote its home services by using a cleaning or house symbol. For Gen Z, who are ever on the go with several responsibilities, these emojis allow users to understand immediately the benefit that the brand offers, appreciating the complexity of the service. These are the advantages that such brands accrue from the use of emojis. They have less competition online with more value deployed to time seeking consumers who want clients to use their services within the shortest time possible.

Telecommunications: Airtel India, Reliance Jio

In the genre of telecommunications, companies such as Airtel India and Reliance Jio offer internet services and mobile data access to millions of subscribers within the country. Such is the growing digitalisation of the country that these brands are a necessary appendage to the daily lives of the Indian consumers, more so, the digitally active Gen Z who are online, be it for studies, work or socializing. For Gen Z, busting the stereotype that emojis are mere decoration of the communication from or about these brands, they are brands: reliability, swiftness and care to the most essential need of the consumers, that of service. Whether it is a smiley face to advocate satisfaction or a lightning bolt advocating speed, these emojis assist brand recall among these young consumers who spend much time using social media and are always looking for quick, precise and friendly responses from the brands. Thus, emoticons have become integral to the brand communication of Airtel and Jio since the younger audience prefers communicating on digital platforms.

Transportation: Olacabs

In India, Olacabs is a famous cab service company that offers its services to a diverse population like young working professionals and students, a large number of whom are Gen Z. In regard to this Schimitar, most Ola tweets involve using Emojis which depict the convenience, safety, and ease of travel, which are core

factors of the younger internet users. Gen Zs who are used to such services being digital decide a more friendly way of communication by implementing use of emojis like the car or map, which makes the brand more appealing. In this manner, brand engagement is maintained even during this offline to online transition through a combination of emoticons in the user interfaces which makes Ola appealing to the users when making a ride request or making an inquiry. This makes the experience better and leaves the consumers wanting for more, especially such that encompasses majority of this critical group of perspective customers.

The selected brands and industries in this study are representative of a wider portion of Indian digital market and economize on ingrain stature of Gen Z as fast-paced, emotionally attached and interactive. The use of emoticons has also learned to brand as a method to win over this particular audience by providing insights that go beyond marketing for the users in all the ways that they wish. Gen Z is of course the most 'digital' generation and has a positive reaction towards the use of emoticons enhancing their relationship with the brand over what the brand is all about. For the primary objective of this research seeks to know the importance and impact of emojis in brand image; As brands operate in different verticals, this research illustrates how these brands can employ the same visual imagery to expand their market reach to younger consumers. Emojis help the users translate their messages through brands hence enhancing the effectiveness of communication in the because the situation demands it. This not only improves the customer experience but deepens their brand loyalty among Gen Z who are used to day-to-day interactions with brands that do not only vie to sell products but also strive to be present, engage, and influence the whole digital sphere.

VADER model-based sentiment analysis

The validity of any sentiment analysis framework is a critical measure, ensuring that the emotion measurement tool accurately captures what it intends to assess. In this research, we employed the VADER (Valence Aware Dictionary and sEntiment Reasoner) model to analyze the sentiment expressed in tweets regarding selected brands. To validate the sentiment scores produced by VADER, we correlated these scores with established positive emoji scores, allowing us to cross-check the effectiveness of our approach.

Both the VADER model is designed to capture positive consumer sentiments and are specifically tailored to the X platform. This compatibility simplifies the correlation process, as the positive sentiment associated with emojis has been documented formany of the brands under examination. By leveraging this existing

knowledge, we can draw meaningful comparisons between the two sentiment measures.

The relationship between VADER sentiment scores is particularly relevant in understanding the emotional landscape surrounding brands. Emojis help figure out what kind of reaction the consumers would have. Even using their feelings, it is thought that VADER sentiment scores will also have positive sentiment. It therefore emphasizes the importance of using VADER as a useful technique in the appraisal of brand engagement and consumer emotions.

This result indicates that the aim of the correlation analysis is to find out whether the VADER model is able to accurately capture the subtle peaks and troughs in the sentiments of the consumers in terms of opinions that have been tweeted out. This assessment reflects VADER's applicatory position in the field of marketing as it is today, making it a critical apparatus in the pursuit of emotional interaction over social networks. By integrating VADER with emoji sentiment measures, we aim to enhance our understanding of how brands can connect with consumers in an increasingly digital marketplace. The Figure 4.1 helps to understand how VADER metrics works to assign the value. Here the compound value is 0.8172 which is relatively a good value, stating the positivity of the text.

To ensure transparency and facilitate replication, the datasets and code used in this study are available in a public repository at https://github.com/DevVora5/Are-Emoji-the-New-Words. The repository includes a sample dataset snippet along with the codec for executing the analysis, enabling independent verification of our approach and supporting further research.

```
>>> from vaderSentiment.vaderSentiment import SentimentIntensityAnalyzer
>>> # Initialize VADER sentiment analyzer
>>> analyzer = SentimentIntensityAnalyzer()
>>> # Example text with emojis
>>> text = "I love this product! 😍👍"
>>> # Analyze sentiment
>>> sentiment = analyzer.polarity_scores(text)
>>> print(sentiment)
{'neg': 0.0, 'neu': 0.516, 'pos': 0.484, 'compound': 0.8172}
>>> |
```

Figure. 1 Image of an example for sentiment analysis on basic most level using the VADER model in python. (Source: Our Study)

FINDINGS

Overview on emojis' usage

The analysis encompassed tweets from various brands, revealing intriguing patterns in emoji usage. A total of 4,631 tweets were analyzed, with 18.74% containing emojis. Notably, a significant proportion of tweets included a single emoji, indicating thatconsumers often rely on one symbol to convey their sentiments. For instance, 13.56% of tweets featured one emoji, while 5.18% used multiple emojis, highlighting a diverse range of expression. The data illustrates that emojis are commonly used alongside text, rather than replacing it entirely, as evidenced by the 81.26% of tweets that contained only text.

The VADER metric used for brands

The VADER metric was employed to gauge consumer sentiment across brands. Table 2 displays the metric values for each of the analyzed brands, ranging from a low of -0.165 for Flipkart to a high of 0.279 for Tanishq Jewelry. The average score across brands was 0.073 with a standard deviation of 0.117. This metric provides a nuanced understanding of how emojis influence brand perception. For instance, 16.56 % of tweets related to Tanishq Jewelry included emojis, significantly impacting its consumer sentiment score.

Using the VADER metric to rank brands

The VADER metric facilitated the ranking of brands based on consumer sentiment. As shown in Table 3, Tanishq Jewelry emerged at the top of the rankings, followed closely by Air Vistara and BigBasket. Conversely, brands like Flipkart and UrbanCompany ranked lower, indicating a correlation between emoji usage and positive sentiment. The findings suggest that brands actively engaging with consumers through emojis tend to foster stronger emotional connections.

Findings from the sentiment analysis using the VADER

The analysis elicited in the research uncovered a range of feelings concerning every brand that corresponded with the consumers' perceptions. Tanishq Jewelry and Amul are among those brands that enjoyed a positive sentiment, as indicated by the average compound score which is one of the measures of sentiment. On the other hand, the Flipkart brands generated negative sentiment suggesting a gap between the messages and strategies being used by the brand and the consumers' expectations.

Of particular interest is the relationship that has been established between the average compound score and average emoji count. This indicates that demographic groups that adopt the use of emojis in their communications with the audience are likely to be more positively perceived by the audience. It seems that the use of emojis, as a nonverbal medium, increases emotional engagement and invokes empathy from the consumers making them have a better consumer experience.

In addition to the primary relationship observed between emoji usage and consumer sentiment, it is important to consider that cultural and industry-specific factors may also be influencing these dynamics. In a diverse country like India, where multiple languages and cultural norms coexist, the interpretation and emotional impact of specific emojis can vary significantly across regions and demographic groups. Forinstance, while universally recognized symbols such as the ☺ (smiley face) and ♡ (heart) consistently convey positive emotions across cultures, an emoji like 🍕 (pizza) may carry additional, context-specific meaning in sectors like food delivery. This cultural nuance means that an emoji conveying warmth or humor in one context might be perceived differently in another.

Moreover, industry-specific communication styles play a crucial role; sectors that rely on informal, playful engagement (such as food delivery or fashion) tend to use emojis in a way that enhances consumer relatability and emotional connection, while more formal sectors (such as telecommunications or finance) might adopt a more restrained approach. Recognizing these alternative explanations not only deepens our understanding of the observed sentiment patterns but also suggests that the impact of emoji usage is multifaceted, with cultural and industry contexts serving as important moderating variables.

Furthermore, the more frequent tweets without any emoji usage tended to have lower compound scores or negative ones. This point demonstrates the necessity to use such an engagement as emoticons in building the brand reputation. The absence of emoticons may lead to emotions being understated missing the purpose of the branding message. The research sheds light upon the changing dynamics of communication with respect to consumers and the brands especially in the context of modern days where everything has become digitized. To have a more coordinated analysis of these phenomena, readers are suggested to consult Appendix C, which provides necessary details on the data and the methodology used in the analysis.

While analyzing the brand performance of Tata associated brands such as Tanishq and Air Vistara, it is apparent that the performance sentiment scores were relatively

higher than those of their competitors. This leaves a very strong brand equity among the users, indicating that Tata's branding construction is very much in tune with the users' expectations. These particular branding messages probably meet brand image requirements since these messages contain the ideas of warranty, quality and "the customer is always right" practices that create positive brand associations in the minds of the consumers. This accounts for the successful interaction of branding, sentiment of the consumers and use of latest technologies in this area, which is also a noteworthy example.

In contrast, e-commerce brands like Flipkart and TataCLiQ perform poorly on the VADER sentiment analysis. This lower performance may stem from consumers frequently expressing dissatisfaction, even over minor issues related to product deliveries. Unlike Tata's luxury and service brands, e-commerce platforms often face challenges due to the immediacy of customer expectations, impacting their overall sentiment scores.

Table 1. Overview of Emoji Usage in Analyzed Tweets (Refer Appendix B for Company wise detailed data.) (Source: Our Study)

	Frequency	Percentage
Tweets only Text	*3763*	*81.26 %*
Tweets combining emojis and Text	*868*	*18.74 %*
Emoji Count in tweets		
1	*628*	*13.56 %*
2	*152*	*3.28 %*
3	*78*	*1.68 %*
4	*8*	*0.17%*
5 and more	*2*	*0.04%*

Table 2. Average Scores and Emoji Usage by Brands (Source: Our Study)

Industry	Brands	Average Score	Average number of emojis in a tweet
Aviation	Air Vistara	0.271	0.24
Dairy	Amul	0.16	0.134
E-Commerce	Flipkart	-0.165	0.316
	Myntra	0.081	0.393
	Tata CliQ	-0.047	0.123
Education & Technology	BYJUS	0.025	0.195
	CRED	0.127	0.149
Entertainment	Dream11	0.181	0.966
Fashion & Lifestyle	Lenskart	0.039	0.079
	Tanishq	0.279	0.205
	Nykaa	0.079	0.152
Fintech	Paytm	0.05	0.74
Food Delivery	Swiggy	0.134	0.186
	Zomato	0.11	0.287
Retail & Grocery	BigBasket	0.186	0.312
	UrbanCompany	-0.044	0.113
Telecommunications	Airtel India	0.054	0.119
	Reliance Jio	-0.014	0.208
Transportation	Ola Cabs	-0.127	0.089

Table 3: Ranking of Brands based on their Scores (Source: Our Study)

Brands	Average Score	Rank
Tanishq	0.279	**1**
Air Vistara	0.271	2
BigBasket	0.186	3
Dream11	0.181	4
Amul	0.160	5
Swiggy	0.134	6
CRED	0.127	7
Zomato	0.110	8
Myntra	0.081	9
Nykaa	0.079	10
Airtel India	0.054	11
Paytm	0.050	12
Lenskart	0.039	13
BYJUS	0.025	14
Reliance Jio	-0.014	15
UrbanCompany	-0.044	16
Tata CliQ	-0.047	17
Ola Cabs	-0.127	18
Flipkart	-0.165	19

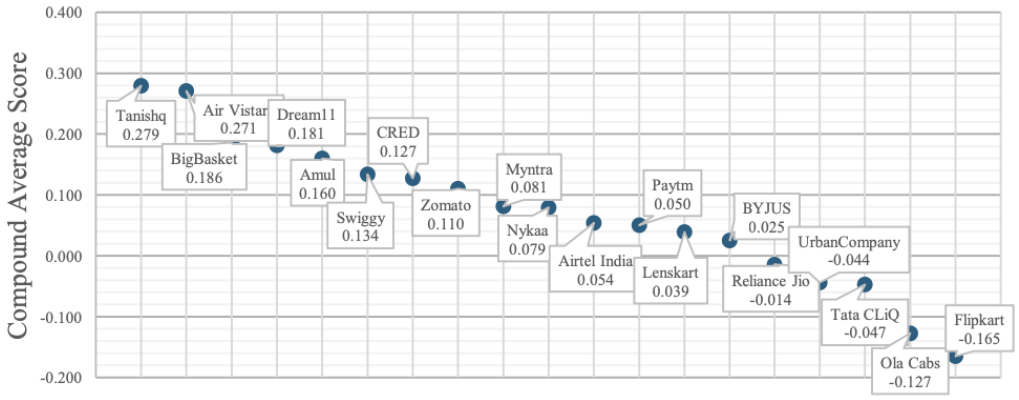

Figure 2: Visualization of the Rankings (Source: Our Study)

IMPLICATIONS

The conclusions extracted from this research, especially when the VADER sentiment analysis tool is used, have far-reaching consequences in terms of management, the research, and the theories employed in marketing and communication.

Managerial Implications

This research is of great importance in view of the constructive aspect concerning managers. The VADER sentiment analysis library available in Python is probably one of the most embedded tools that even nontechnical subjects too can successfully conduct both basic and advanced consumer sentiment analysis. This is a library that even persons with lithe programming skills can apply and study consumer sentiment. This advantage makes it possible for companies to be able to quickly analyse and respond to the views of the consumers, which is something very essential today where consumer taste tends to change quickly.

It is through a sentiment analysis of the products that refinement of product development processes can be achieved. For example, where a brand does a VADER and determines that consumers express being excited about a particular product feature or experience, then the bearing of the consumers' opinion on future products can be managed. Such responsiveness improves the speed of the innovation process as well as markets the product more to the consumers due to a higher accuracy of addressing the needs.

Moreover, the inclusion of both dominance in VADER analyzing features such as emojis and messages broadens the discussions on consumer sentiments analysis. Typical measures or evaluation methods are limited to textual or visuals and do not give attention on emotions depicted by the use of emojis. This special trait enables managers to gather diverse emotions from consumers which are useful in developing better marketing approaches. For instance, when VADER specifies that specific emojis are positively related to consumer feelings towards a brand, management's strategies can use these emotions in ads and touch consumer sentiments more in their add campaigns.

Equally beneficial, this strategy in sentiment analysis helps managers in their advertising initiatives to prevent miscommunication and overexposure. They comprehend the role of emojis, in seeing to it that the opposite occurs, that is, the emotions in the text are even amplified or simply remain the same, and hence ensure that texts that will accompany the advertisements are within the scope of

the intended target audience. Such type of marketing increases engagement but also creates greater relationships with the customers since they feel appreciated and understood.

To conclude, it is evident that the results obtained from sentiment analysis using VADER will be beneficial to the company in question. This instrument promotes a timely change in product design and marketing tactics that align with the needs of consumers. In the end, an organization's ability to measure and meet consumer needs and wishes as they arise has great potential from the point of view of customer satisfaction and brand loyalty as well as business performance.

Researcher Implications

There are practical lessons to be learned from this study that go beyond the findings especially in terms of research methods and strategies. A simple to understand and consider a framework for evaluating consumer sentiment is anchored on the VADER (Valence Aware Dictionary and sEntiment Reasoner) model. Thus, other than endorsement, this foreign model is cheap and has available literature supporting its implementation. This would encourage scholars to focus on consumer sentiment because it is easy, hence broadening understanding of the interplay of emotions in brand interaction.

Furthermore, VADER is quite flexible as people can analyze a dataset from different platforms without the need of extensive preparation which is usually done beforehand during general sentiment analysis tasks. Consequently, they have the opportunity to study the consumer's attitude in different situations which broadens the scope of stance towards the impact of communication on the behavior of consumers. Of course, including emoji in the study of how consumers impact corporate communication management correspond with Social Presence Theory which claims that the knowledge of people in communication influences feelings (Short et al, 1976). Since emojis help establish emotional ties between companies and consumers, they are essential in communication via the internet.

Theoretical Implications

Furthermore, this research has made a considerable amount of theoretical contribution, specifically when it comes to considering emotions through both the words and the pictures. While such an investigation confines the dominance of the drawing languages narrative-codes approach through their interaction with words, their image-words association should not also remain in isolation within the two communicative aspects. This aspect is pertinent in the current era especially since

it is common that consumers focus on visual representations of the products or exchange information through pictorial depictions.

Results also demonstrate shifts with respect to the attitudes towards brands and how the activity encourages the audience to take up. When emojis become woven into the social fabric, there arises an urgency to redefine the construction of the ideation of emotionality and perception of the brand. This study also has a nexus with Uses and Gratifications Theory which hold that individuals act on the media to gratify some emotions (Katz et al 1973). Emojis help meet these objectives by allowing for communication and interaction and are therefore beneficial to the consumer.

In addition, the ELM model also explains the use of emojis which Petty and Cacioppo(1986) defines as persuading the recipient through peripheral cues without conscious thought processes. This emphasizes the importance of visual communication to be integrated into branding since it portrays and influences the brand's position on the market. It further utilizes Affect Theory that studies emotions in consumers, proposes that emojis improve emotions towards a brand and determine brand attitude (Russell, 1980).

The research also contributes to Brand Equity Theory, highlighting how positive sentiments fostered through emojis can enhance brand value and consumer relationships (Aaker, 1991). Similarly, it engages with semiotics and visual communication, illustrating how emojis act as symbols that enrich brand messaging (Barthes, 1964).

Considering the Digital Marketing Communication Theory, an analysis of tweets containing emoji used in brands can help to find out how images affect the communicated brand (Chaffey & Ellis-Chadwick, 2019). At last, the application of Cognitive Dissonance theory provides explanation of how emojis help to reduce this negative response by enhancing positive attitudes towards a brand (Festinger, 1957).

All in all, this paper brings forth excellent contribution to management, research and theory. It bridges the widening denominator between text based and visual sentiment analysis making way for fresh research and practice in marketing and communication studies. The combination of emojis and sentiment analysis does not only give us more insight about consumers' behaviors but also contributes more to the direction of further studies in the era where everything is electronically connected.

LIMITATIONS

Some impressive accomplishments have been made by this study in the area of sentiment analysis where the VADER model was developed; however, some remarkable limitations should be highlighted. Such limitations may owe to the constituent features of natural language processing systems, the functionality of the VADER system itself in this matter, the strategy that was used to collect information, and the limits or possible parameters of the findings.

Due to its focus on informal syntax, abbreviations, and the use of emoticons which is very significant in the analysis of social networks, VADER is unable to discern sarcasm, humor, irony, and other hard nuances of language. Communication is also done more effectively using these methods, especially when the consumer is annoyed with the brand or attempts in great humor using the brand. With such constraint, VADER may miss out on these nuances and thereby misinterpret sentiment and that will, in the end, skew the analysis. For example, when a person makes a sarcastic remark, the person's entire feeling of that remark may be diametrically opposed to how one interprets it based on some tweets for instance. There are tweets which arenot correctly classified by VADER because VADER interprets sarcastic tweets as neutral or positive and does not treat them as negative comments and that mostly affects the extent of VADER's correctness in certain aspects.

Moreover, there's yet another inadequacy of VADER, which is the limited range of emojis that are stored in their databases. This is due to emoji recognition and processing capability of only 625 which does not even constitute 1% of the current total of around 3570 emojis. In written communication, the use of symbols is often used more often since they symbolize emotions rather than words themselves. To illustrate this, by ignoring a large volume of emoji count Monsieur Icono, primarily newer or infrequently use icons would create a handicap to model because certain clues would be missing, which would render the emotion analysis ineffective. This centered the users' attention on the language and this void could be one of the constraints to the VADER's performance on account of the effectiveness of online generated images that have been subject to analysis.

Furthermore, as people engage in data collection, they have an inclination to display bias and especially the X handles of brands. Evidence of this claim can be seen in the results which clearly suggest that, brand handles on X are most fruitful in obtaining those tweets that are directed towards customer care agents for issues around complaints or complaint requests, thus complaint interactions with

consumers. Therefore, the dataset is expected to put a lot of weight on the marketing aspects of the various interactions and very little on the conversational or marketing or positive brand attachments that come from a more organic endorsement of a product or its knowledge as part of one's lifestyle. Thus, it may be controversial to state the less sentient atmosphere present analysis of the results, emphasizing the one-sided "negative" affiliate of the marketed, consumer's, cognitive-emotional spectrum of feelings. X also does not desire accurately articulate bikes emotions because rapid messages with adaptations that are only typical or relevant at that period are composed. Neither does X company history tell how people feel since they answer over emotions in phases high-level bursts for the time limit present only those feelings appropriate for that time.

While this dataset mostly covers X information, it can also be limiting in some areas. Although X is a useful tool for brand interaction, it only presents an opportunity to view brand engagement activities. People communicate and interact with various platforms using different modes and tones. For instance, Instagram supports more imagery and storytelling, while Reddit is more interaction and discussion based. Therefore, the sentiments that are extracted through X's data, for instance, may not be applicable or seen on such sites that have other means of seeking the consumers' feedbacks.

Also, it must be said that the geographical scope of the research as implying a selection of popular brands well known to the appropriate audience within more than one industry has evident consequences concerning the extendibility of the findings. Incertain situations, especially some of the relationships will complement behavior toward those sectors that are characterized by an operating economy of contraction. Specifically, both will avoid any references to such markets where the consumers are not active or engaged and are not willing to ventilate and pursue engagement with brands in public, such as B2B or niche areas. Furthermore, cross-cultural characteristics to some extent are determinants of how and what online consumers 'feel', in particular, what kind of language and emotion symbols are adopted.

While our study offers valuable insights into the impact of emojis on consumer sentiment within the Indian market, it is important to note that these findings are deeply rooted in India's unique socio-cultural and digital landscape. The selected brands, data collection methods, and communication styles reflect the distinctive characteristics of Indian consumer behavior and may not directly translate to other cultural or linguistic contexts. Consequently, caution should be exercised when generalizing these results beyond India. Future research could benefit from cross-

cultural studies to explore whether similar patterns of emoji usage and consumer sentiment are observed in different markets.

However, across borders we tend to understand specific aspects, communicated through a certain way, differently which works against the use of these findings in cases where there is a shift from one segment to another. Even with these shortcomings, this study is a significant step forward in sentiment analysis by the application of VADER in the real world. In spite of this, recommendations are made that future studies will seek to detect adaptive sentiments more accurately than this study in light of these limitations by complex language enhancements of the model and expanding the emoji collection as well as getting information from other than one platform. If this is the case, studies on satirism as a genre will advance to more positive ones concerning the studying of shifting brand-consumer relationships to a qualitatively new level.

SCOPE FOR FURTHER RESEARCH

While this manuscript has been successful in its contribution to using VADER Model in information mining or more particularly in brands and tweets consumer behaviour aspects, there are many gaps that deserve to be filled. These areas concern improving such models, broadening the sample population, collecting perception data across platforms, and providing cultural context.

Another aspect that would need further exploration is what can be termed as improving models used in the analysis of sentiments. Despite the fact that VADER has a natural head for language that is a bit more emotional than formal, in terms of typical social media discourse with emojis and infographics, the framework also has its obvious drawbacks for example with sarcasm or even contextual meanings. On the other hand, the languages such as NLU or deep learning could be incorporated with the VADER models to create hybrids. More specifically, however, text and contextof user activities and even visual images promoting subjects that are measured by their popularity could be focused on in the future models. Emprical language models such as BERT Vice versa would enhancing such language models by utilising models that are able to read emotion in language and socia context and thus translating them back into the socia language would benefit the models where language and emotion comprehension model knowledge is needed.

Another area for future research includes enlarging the scope of the study into dimensions outside the X site. X is indeed a very valid place to analyze the views of consumers, however this is but one aspect of the entire interaction. It would be worth investigating additional platforms, such as Instagram, Facebook, and Reddit, as well as video platforms, such as Tiktok or Youtube in which consumer sentiment can be expressed in various formats and visuals. Each channel has also its type of interaction and audience, which can determine how emotions are portrayed. Because different channels will be investigated in the future research, it will be assumed that understanding consumer emotions in relation to products and brands will be more sensibly developed.

Due to their varying styles, many will tackle different angles on the same central theme. Comments could extend this work by examining emotions connected to different means of communication that can play a role in performing the act of emotion: emotion in text, image, video, or meme. When producing video materials for social networks such as Instagram and TikTok, users are likely/will seek to interact with the content in a manner that does not necessarily eulogize text-based emotion serving models, the ease with which visual-based contents can be deployed is very ench. Restricting research that eliminates the use of image understanding, video understanding or meme based sentiment analysis techniques can restrict the full potential reserved for sentiment analysis. It will further facilitate the rise of more complex computational multimodal sentiment analysis approaches such as content based image/text sentiment analysis more relevant to the current day practice of marketing communications.

In the same context, more research in the future can seek to understand how each of the culture and region influences the use of sentiments. In this regard, use of emojis is expected to make sense only with reference to a culture as they may be context based emotions. For instance, I took this forward and put an image of a smiley which is acceptable in the US culture in contrast to other cultures where sometimes non verbal communications are not regarded as appropriate. Of note in this context, it would be fascinating to study a certain cross cultural consumer segment and their feelings and the way they show them in the internet. This is particularly relevant in the communication of global corporations who have presence in different countries and engage in promotion and sales in many cultural environments.

Yet other possible future research directions may include grappling with the effect of the fluid component – that is, the sphere of the interaction itself. Moreover, due to thechanging dynamics of time and introduction of new technologies such as ai

chatbots, virtual reality, application software, the end users' engagements with the branded products are expected to change. According to the future studies, new content strategies aimed at consumer user research metrics focused particularly on sentiment analysis, VADER in this context, may be applied and require transformations in current and future formats. For example, more advanced techniques for understanding appreciation, such as through phone conversations or using augmented reality, may be indispensable as these practices become prevalent in brand-consumer interactions.

Finally, some further estimation efforts may provide a redefine scope of product sentiment magnitudes by introducing something like sub-indices that may adequately capture peculiarities of niche, type or industries. This study used well known B2C brands, however in the future it is possible that research on the sentiment analysis will focus on B2B or specific industry such as hospitality or health care, which are both consumer opinion based. Such models of sentiment analysis, when used in accordance with the specific terminologies, communication patterns and needs of various industries, shall improve the usability of these solutions and provide more useful recommendations to the brands.

In conclusion, while this study has showed the usability of the VADER to some extent, there are plenty of other areas yet to be thoroughly researched. Future other research on this subject area can address the current shortcomings by improving on model efficacy, using wider data sets and comprehending culture and context in sentiment better. In light of these findings, a more rigorous approach is needed to assess whether emoji usage significantly predicts consumer sentiment. By employing advanced statistical methods—such as multivariate regression—future research can account for variables like tweet length, inherent text sentiment, and industry-specific communication styles. This strategy would help isolate the unique influence of emojis on sentiment scores, thereby strengthening our causal inferences and enhancing our understanding of how these visual cues shape consumer-brand interactions. Overall, future works can build on existing efforts in this field and provide added value to information/image practitioners about consumer insight more effectively in view of the increasing intricacies in the digital environment.

CONCLUSION

The findings of this study provide clear evidence that emoji usage in brand-related tweets significantly influences consumer sentiment and brand equity in the Indian market. Tweets containing positive emojis generally exhibit higher sentiment

scores, reinforcing the idea that emojis help create emotional connections with consumers. The context in which emojis are used also plays a role—brands that incorporate emojis effectively in promotional and engagement-focused interactions tend to foster stronger consumer connections compared to those that use them sparingly or in customer service-related contexts. Additionally, a higher frequency of emoji usage iscorrelated with increased engagement, indicating that emojis serve as a valuable tool for brands aiming to enhance consumer interaction and brand perception on social media. These findings confirm that emojis are more than just visual elements; they are strategic assets in digital marketing that influence brand equity and consumer sentiment.

This research explores how emotion is expressed through the use of emojis within different brands in India and provides useful perspectives on the dominant new ways of communication. In the introduction, we established the premise that emojis are extremely effective especially for young people such as Gen Z, who are more exposed to digital communications, in augmenting emotions.

The literature review, however, brought forth inadequacies in the study of the effects of visual communication with the audience on the brand and introduces our research projects. Our construing of the VADER metric caused the disentanglement of the assessment of the attitudes of customers towards the products strategically opening avenues to measure the extent of the impact of emoji usage on the level of engagement with the brand.

Through an appropriate methodology, we researched a wide range of tweets and identified that there are different patterns of emoji usage across the different brands. It was further found that the brands such as Tanishq Jewelry and Amul which used emoji's effectively had higher average compound scores which meant that there was some positive relationship between the usage of emoji's and the attitudes of consumers. To be precise, as a result of Tanishq's use of emojis, the company was credited with a sentiment score of 0.279, which goes to show that emojis improve the perception of the brand and not only appealing to the judges. However, Tanishq and Ola Cabs suffered from negative sentiment because of the low engagement in emoji's and that is backward.

These findings have bear more implications for advertisers and brand custodians. They assert the importance of including emojis in the communication plans so that there are emotional concerns towards the product and its endorsement. The present research depicts that consumers have a positive attitude to brand messages when such messages are enhanced through the use of emojis as a way of humanization

of brand which could lead to emotional affection and subsequent brand attachment. In addition, the research brings forth the essence of possession of brand reputation as seen with endorsers associated with the Tata firm that received higher sentiment scores and sentiments from the public. This particular research implies that, strong brands tend to abuse emojis by improving brand interest and engagement further. Though this research made contributions as elucidated above, it was not free from limitations. The emphasis on tweets may limit the level at which such findings can be replicated on other social networking sites or other modes of communication. Also, the changing use of emojis calls for further studies to keep track of the changing consumer behavior patterns as well as the socio-cultural environment.

In future studies, it would be apt to delve into the contextual determinants of emoji use and consider how suitably emoji interpretation and effects may differ across people and places. Also, addressing the follow-up issues regarding the contribution of emoji engagement to the brand over time and how these strategies can be used in the effective communication of brands will add more in this discussion. This study, therefore, addresses the role of emojis in the current marketing strategies, and furthermore, suggests that marketers should change their communication strategies in the new age which is highly visual, to gain better access to the target audience.

This study confirms that emojis play a crucial role in shaping consumer sentiment and brand equity in social media conversations. By analyzing sentiment scores and engagement levels across multiple industries, we demonstrate that emojis enhance brand communication and consumer interaction. These findings align with the study's initial objectives, reinforcing that visual symbols are not merely decorative but serve as valuable tools for digital marketing and brand strategy in the Indian market.

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APPENDIX - A

Tweets Collected from Selected Brands (Source: Our Study)

Company	Tweets Collected
Airtel India	278
Air Vistara	196
Amul	307
BigBasket	154
BYJUS	133
CRED	148
Dream11	208
Flipkart	269
Lenskart	127
Myntra	191
Nykaa	132
Ola Cabs	179
Paytm	219
Reliance JIO	274
Swiggy	1000
Tanishq	151
Tata CLiQ	130
UrbanCompany	124
Zomato	411
Total	4631

This table provides a detailed overview of the number of tweets collected for each brand in the study. The data highlights the varying levels of engagement across different brands, with a total of 4631 tweets analyzed for sentiment related to consumer interactions, average of approx. 250 per brand.

APPENDIX - B

Company-wise Breakdown of Tweets Containing Emojis (Source: Our Study)

Company	Tweets with Emoji Count 0	Tweets with Emoji Count 1	Tweets with Emoji Count 2	Tweets with Emoji Count 3	Tweets with Emoji Count 4	Tweets with Emoji Count more than 4
Airtel India	251	23	3	0	1	0
Air Vistara	151	43	2	0	0	0
Amul	269	35	3	0	0	0
BigBasket	123	20	7	2	2	0
BYJUS	115	13	2	3	0	0
CRED	132	12	3	0	1	0
Dream11	115	30	19	43	1	0
Flipkart	208	44	13	2	1	1
Lenskart	118	8	1	0	0	0
Myntra	136	39	12	4	0	0
Nykaa	119	7	5	1	0	0
Ola Cabs	167	9	2	1	0	0
Paytm	123	49	31	14	1	1
Reliance JIO	230	34	8	1	1	0
Swiggy	838	141	18	3	0	0
Tanishq	126	20	4	1	0	0
Tata CLiQ	119	7	3	1	0	0
UrbanCompany	113	8	3	0	0	0
Zomato	310	86	13	2	0	0
Total	3763	628	152	78	8	2

This table presents the distribution of tweets with varying counts of emojis for each analyzed brand. It highlights how many tweets contained zero, one, two, three, four, or more than four emojis, providing insights into emoji usage patterns across different companies.

APPENDIX - C

Average Sentiment Scores for Brands with and without Emoji Usage (Source: Our Study)

Brands	Average Score	Average Score for Tweets with Emojis	Average Score for Tweets without Emojis
Airtel India	0.054	0.349	0.022
Air Vistara	0.271	0.445	0.219
Amul	0.160	0.307	0.139
BigBasket	0.186	0.392	0.134
BYJUS	0.025	0.217	-0.005
CRED	0.127	0.593	0.071
Dream11	0.181	0.185	0.178
Flipkart	-0.165	0.003	-0.215
Lenskart	0.039	0.059	0.037
Myntra	0.081	0.263	0.008
Nykaa	0.079	0.410	0.043
Ola Cabs	-0.127	0.417	-0.166
Paytm	0.050	0.058	0.045
Reliance Jio	-0.014	0.103	-0.037
Swiggy	0.134	0.205	0.120
Tanishq	0.279	0.588	0.218
Tata CLiQ	-0.047	0.217	-0.072
UrbanCompany	-0.044	0.413	-0.088
Zomato	0.110	0.192	0.084

This table presents the average sentiment scores for various brands, comparing the overall score with the scores derived from tweets containing emojis and those without. The data highlights the impact of emoji engagement on consumer sentiment, illustrating how brands like Tanishq and Air Vistara achieve significantly higher scores in tweets that include emojis, while brands such as Flipkart and Ola Cabs show contrasting trends in sentiment when emojis are absent.